

Analysis of EMI DDI, and TPL series

Segun Aroyehun

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(zoo)
```

```
##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
library(Hmisc)
```

```
##
## Attaching package: 'Hmisc'

## The following objects are masked from 'package:dplyr':
##
##   src, summarize

## The following objects are masked from 'package:base':
##
##   format.pval, units
```

```
library(lmtest)
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

```
library(ggplot2)
library(ggrepel)
library(stargazer)
```

```
##
## Please cite as:

## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##      recode
```

```
library(sandwich)
library(clubSandwich)
```

```
## Registered S3 method overwritten by 'clubSandwich':
##      method      from
##      bread.mlm sandwich
```

```
library(ggeffects)
library(margins)
library(marginaleffects)
library(boot)
```

```
##
## Attaching package: 'boot'

## The following object is masked from 'package:car':
##
##      logit
```

```
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##      group_rows
```

```
library(modelsummary)
```

```
##  
## Attaching package: 'modelsummary'  
  
## The following object is masked from 'package:Hmisc':  
##  
##      Mean
```

```
library(visreg)  
library(tinytable)  
set.seed(1234)
```

```
output_path <- "./output"  
f <- function(x) format(x, digits = 3, nsmall = 2)  
df_us <- read.csv("../aggregate_data/US/agg_sbertllms_gt1945.csv")  
df_de_west <- read.csv("../aggregate_data/DE/agg_sbertllms_lt1990.csv")  
df_de_ge1990 <- read.csv("../aggregate_data/DE/agg_sbertllms_ge1990.csv")  
df_it <- read.csv("../aggregate_data/IT/agg_sbertllms_gt1945_attributed.csv")  
df_is <- read.csv("../aggregate_data/IS/agg_sbertllms_gt1945.csv")  
df_pl <- read.csv("../aggregate_data/PL/agg_sbertllms_gt1989.csv")  
df_tr <- read.csv("../aggregate_data/TR/agg_sbertllms_gt1981_attributed.csv")
```

```
df_us <- df_us |>  
  arrange(year, .by_group = TRUE) |>  
  mutate(  
    period_index = row_number(),  
    emi = evidence_intuition_diffpool_score_mean,  
    ddi_lag1 = dplyr::lag(ddi, 1),  
    emi_lag1 = dplyr::lag(emi, 1),  
    tpl_lag1 = dplyr::lag(transparent_laws, 1),  
    log_gdp = log(gdp_per_capita)  
  ) |>  
  ungroup()
```

```
df_de_ge1990 <- df_de_ge1990 |>  
  arrange(year, .by_group = TRUE) |>  
  mutate(  
    period_index = row_number(),  
    emi = evidence_intuition_diffpool_score_mean,  
    ddi_lag1 = dplyr::lag(ddi, 1),  
    emi_lag1 = dplyr::lag(emi, 1),  
    tpl_lag1 = dplyr::lag(transparent_laws, 1),  
    log_gdp = log(gdp_per_capita)  
  ) |>  
  ungroup()
```

```
df_de_west <- df_de_west |>  
  arrange(year, .by_group = TRUE) |>  
  mutate(  
    period_index = row_number(),  
    emi = evidence_intuition_diffpool_score_mean,  
    ddi_lag1 = dplyr::lag(ddi, 1),
```

```

    emi_lag1 = dplyr::lag(emi, 1),
    tpl_lag1 = dplyr::lag(transparent_laws, 1),
    log_gdp = log(gdp_per_capita)
  ) |>
  ungroup()

df_it <- df_it |>
  arrange(year, .by_group = TRUE) |>
  mutate(
    period_index = row_number(),
    emi = evidence_intuition_diffpool_score_mean,
    ddi_lag1 = dplyr::lag(ddi, 1),
    emi_lag1 = dplyr::lag(emi, 1),
    tpl_lag1 = dplyr::lag(transparent_laws, 1),
    log_gdp = log(gdp_per_capita)
  ) |>
  ungroup()

df_is <- df_is |>
  arrange(year, .by_group = TRUE) |>
  mutate(
    period_index = row_number(),
    emi = evidence_intuition_diffpool_score_mean,
    ddi_lag1 = dplyr::lag(ddi, 1),
    emi_lag1 = dplyr::lag(emi, 1),
    tpl_lag1 = dplyr::lag(transparent_laws, 1),
    log_gdp = log(gdp_per_capita)
  ) |>
  ungroup()

df_pl <- df_pl |>
  arrange(year, .by_group = TRUE) |>
  mutate(
    period_index = row_number(),
    emi = evidence_intuition_diffpool_score_mean,
    ddi_lag1 = dplyr::lag(ddi, 1),
    emi_lag1 = dplyr::lag(emi, 1),
    tpl_lag1 = dplyr::lag(transparent_laws, 1),
    log_gdp = log(gdp_per_capita)
  ) |>
  ungroup()

df_tr <- df_tr |>
  arrange(year, .by_group = TRUE) |>
  mutate(
    period_index = row_number(),
    emi = evidence_intuition_diffpool_score_mean,
    ddi_lag1 = dplyr::lag(ddi, 1),
    emi_lag1 = dplyr::lag(emi, 1),
    tpl_lag1 = dplyr::lag(transparent_laws, 1),
    log_gdp = log(gdp_per_capita)
  ) |>
  ungroup()

```

```
cor.test(df_us$emi, df_us$ddi)
```

```
##  
## Pearson's product-moment correlation  
##  
## data: df_us$emi and df_us$ddi  
## t = 3.6626, df = 78, p-value = 0.0004529  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## 0.1783709 0.5559946  
## sample estimates:  
## cor  
## 0.3830747
```

```
cor.test(df_de_ge1990$emi, df_de_ge1990$ddi)
```

```
##  
## Pearson's product-moment correlation  
##  
## data: df_de_ge1990$emi and df_de_ge1990$ddi  
## t = 3.544, df = 30, p-value = 0.001314  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## 0.2400134 0.7498887  
## sample estimates:  
## cor  
## 0.5432427
```

```
cor.test(df_de_west$emi, df_de_west$ddi)
```

```
##  
## Pearson's product-moment correlation  
##  
## data: df_de_west$emi and df_de_west$ddi  
## t = 2.4431, df = 39, p-value = 0.01919  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## 0.0638246 0.6042457  
## sample estimates:  
## cor  
## 0.3643214
```

```
cor.test(df_it$emi, df_it$ddi)
```

```
##  
## Pearson's product-moment correlation  
##  
## data: df_it$emi and df_it$ddi  
## t = 4.1146, df = 75, p-value = 9.847e-05  
## alternative hypothesis: true correlation is not equal to 0
```

```
## 95 percent confidence interval:
## 0.2269786 0.5958475
## sample estimates:
##      cor
## 0.4291406
```

```
cor.test(df_is$emi, df_is$ddi)
```

```
##
## Pearson's product-moment correlation
##
## data: df_is$emi and df_is$ddi
## t = 5.1042, df = 76, p-value = 2.383e-06
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3185471 0.6542329
## sample estimates:
##      cor
## 0.5052616
```

```
cor.test(df_pl$emi, df_pl$ddi)
```

```
##
## Pearson's product-moment correlation
##
## data: df_pl$emi and df_pl$ddi
## t = 3.3547, df = 34, p-value = 0.001964
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.2034792 0.7107665
## sample estimates:
##      cor
## 0.4986782
```

```
cor.test(df_tr$emi, df_tr$ddi)
```

```
##
## Pearson's product-moment correlation
##
## data: df_tr$emi and df_tr$ddi
## t = 4.6096, df = 39, p-value = 4.257e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.3501986 0.7622512
## sample estimates:
##      cor
## 0.5938666
```

```
# pooled
df_all <- bind_rows(
  "United States" = df_us,
```

```

"Italy" = df_it,
"Germany" = df_de_ge1990,
"West Germany" = df_de_west,
"Iceland" = df_is,
"Poland" = df_pl,
"Turkey" = df_tr,
.id = "country"
)
df_all$clientelism_rev <- -df_all$clientelism_index
cor.test(df_all$emi, df_all$ddi)

```

```

##
## Pearson's product-moment correlation
##
## data: df_all$emi and df_all$ddi
## t = 4.1326, df = 383, p-value = 4.409e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.1089131 0.3003533
## sample estimates:
## cor
## 0.2066098

```

```

#plotting theme
theme_house <- theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor = element_blank(),
    plot.title = element_text(face = "bold"),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10)
  )
theme_set(theme_house)

#model output format
f_est <- function(x) formatC(x, format="f", digits=3)
f_p <- function(x) {
  ifelse(
    x < 0.001,
    formatC(x, format = "e", digits = 3),
    formatC(x, format = "f", digits = 3)
  )
}

```

```

df_all$country <- factor(df_all$country)
df_all$country <- relevel(df_all$country, ref = "United States")
df_all$year <- factor(df_all$year)

emi_model <- lm(emi ~ year + ddi + country, data = df_all)
summary(emi_model)

```

```

##
## Call:

```

```
## lm(formula = emi ~ year + ddi + country, data = df_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.245125 -0.051283  0.002797  0.055692  0.215106
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.434784    0.064949  -6.694 1.08e-10 ***
## year1947      -0.016081    0.075095  -0.214 0.830586
## year1948       0.111085    0.075098   1.479 0.140146
## year1949       0.006449    0.070517   0.091 0.927190
## year1950       0.057106    0.070627   0.809 0.419417
## year1951       0.061353    0.070669   0.868 0.385997
## year1952       0.100753    0.070634   1.426 0.154793
## year1953       0.051998    0.070682   0.736 0.462519
## year1954       0.149111    0.070814   2.106 0.036069 *
## year1955       0.174699    0.070826   2.467 0.014203 *
## year1956       0.177919    0.070856   2.511 0.012569 *
## year1957       0.186426    0.070879   2.630 0.008977 **
## year1958       0.217098    0.070914   3.061 0.002403 **
## year1959       0.216248    0.070940   3.048 0.002508 **
## year1960       0.202099    0.071088   2.843 0.004779 **
## year1961       0.234995    0.071155   3.303 0.001075 **
## year1962       0.291360    0.071202   4.092 5.51e-05 ***
## year1963       0.319494    0.071276   4.483 1.05e-05 ***
## year1964       0.278688    0.071292   3.909 0.000115 ***
## year1965       0.332198    0.071312   4.658 4.81e-06 ***
## year1966       0.361296    0.071373   5.062 7.27e-07 ***
## year1967       0.345270    0.071407   4.835 2.13e-06 ***
## year1968       0.276563    0.071390   3.874 0.000132 ***
## year1969       0.319743    0.071418   4.477 1.08e-05 ***
## year1970       0.265352    0.071663   3.703 0.000254 ***
## year1971       0.310516    0.071686   4.332 2.02e-05 ***
## year1972       0.292523    0.071732   4.078 5.83e-05 ***
## year1973       0.332486    0.071783   4.632 5.42e-06 ***
## year1974       0.278906    0.071811   3.884 0.000127 ***
## year1975       0.289747    0.071980   4.025 7.22e-05 ***
## year1976       0.286833    0.072121   3.977 8.76e-05 ***
## year1977       0.292148    0.072346   4.038 6.86e-05 ***
## year1978       0.276613    0.072388   3.821 0.000162 ***
## year1979       0.336371    0.072392   4.647 5.07e-06 ***
## year1980       0.307707    0.072671   4.234 3.06e-05 ***
## year1981       0.294359    0.072636   4.053 6.47e-05 ***
## year1982       0.263627    0.068758   3.834 0.000154 ***
## year1983       0.247078    0.069225   3.569 0.000417 ***
## year1984       0.227729    0.070030   3.252 0.001278 **
## year1985       0.242925    0.070030   3.469 0.000600 ***
## year1986       0.218075    0.070087   3.112 0.002042 **
## year1987       0.219447    0.070148   3.128 0.001932 **
## year1988       0.230239    0.070322   3.274 0.001185 **
## year1989       0.277811    0.070304   3.952 9.70e-05 ***
## year1990       0.216418    0.068938   3.139 0.001863 **
## year1991       0.127019    0.069532   1.827 0.068735 .
```

```

## year1992      0.150374    0.069776    2.155 0.031955 *
## year1993      0.151250    0.069909    2.164 0.031295 *
## year1994      0.178565    0.069881    2.555 0.011107 *
## year1995      0.139192    0.069798    1.994 0.047042 *
## year1996      0.150666    0.069853    2.157 0.031814 *
## year1997      0.202824    0.069913    2.901 0.003996 **
## year1998      0.207870    0.069913    2.973 0.003187 **
## year1999      0.208319    0.069946    2.978 0.003137 **
## year2000      0.272295    0.069937    3.893 0.000122 ***
## year2001      0.232931    0.069532    3.350 0.000912 ***
## year2002      0.225721    0.069667    3.240 0.001330 **
## year2003      0.256130    0.069863    3.666 0.000291 ***
## year2004      0.257902    0.069969    3.686 0.000271 ***
## year2005      0.234083    0.069913    3.348 0.000918 ***
## year2006      0.229709    0.069900    3.286 0.001137 **
## year2007      0.241408    0.070058    3.446 0.000651 ***
## year2008      0.291237    0.070062    4.157 4.22e-05 ***
## year2009      0.248645    0.070152    3.544 0.000457 ***
## year2010      0.257400    0.070034    3.675 0.000282 ***
## year2011      0.227236    0.069863    3.253 0.001275 **
## year2012      0.216779    0.069789    3.106 0.002078 **
## year2013      0.194261    0.069153    2.809 0.005295 **
## year2014      0.217274    0.068867    3.155 0.001769 **
## year2015      0.179297    0.068516    2.617 0.009327 **
## year2016      0.202917    0.067600    3.002 0.002912 **
## year2017      0.206861    0.066969    3.089 0.002198 **
## year2018      0.177639    0.066960    2.653 0.008408 **
## year2019      0.209398    0.066902    3.130 0.001922 **
## year2020      0.173814    0.066640    2.608 0.009559 **
## year2021      0.179207    0.066783    2.683 0.007695 **
## year2022      0.225101    0.068506    3.286 0.001138 **
## year2023      0.134343    0.076922    1.746 0.081756 .
## year2024      0.120790    0.086513    1.396 0.163693
## year2025      0.033529    0.085069    0.394 0.693763
## ddi           0.295858    0.076864    3.849 0.000145 ***
## countryGermany -0.035333    0.021493   -1.644 0.101246
## countryIceland -0.013164    0.014656   -0.898 0.369813
## countryItaly    0.026438    0.014742    1.793 0.073920 .
## countryPoland   0.027116    0.020920    1.296 0.195912
## countryTurkey   0.132232    0.038155    3.466 0.000607 ***
## countryWest Germany -0.049883    0.021326   -2.339 0.019992 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0918 on 298 degrees of freedom
## Multiple R-squared:  0.5291, Adjusted R-squared:  0.3932
## F-statistic: 3.893 on 86 and 298 DF,  p-value: < 2.2e-16

```

```
vif(emi_model)
```

```

##           GVIF Df GVIF^(1/(2*Df))
## year      4.119989 79      1.009001
## ddi       7.297413  1      2.701372
## country 10.575439  6      1.217190

```

```
adf.test(residuals(emi_model))
```

```
## Warning in adf.test(residuals(emi_model)): p-value smaller than printed p-value
```

```
##
## Augmented Dickey-Fuller Test
##
## data: residuals(emi_model)
## Dickey-Fuller = -5.612, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

```
kpss.test(residuals(emi_model))
```

```
## Warning in kpss.test(residuals(emi_model)): p-value greater than printed
## p-value
```

```
##
## KPSS Test for Level Stationarity
##
## data: residuals(emi_model)
## KPSS Level = 0.066049, Truncation lag parameter = 5, p-value = 0.1
```

```
jarque.bera.test(residuals(emi_model))
```

```
##
## Jarque Bera Test
##
## data: residuals(emi_model)
## X-squared = 2.0865, df = 2, p-value = 0.3523
```

```
#plot
```

```
p_vis <- visreg(emi_model, "ddi", gg = TRUE, xlab="DDI", ylab="EMI", line.par = list(col = "black"))+
  theme_house +
  scale_color_manual(values = "black") +
  scale_fill_manual(values = "grey70")
```

```
ggsave(filename = file.path(output_path, "EMI-DDI-model.pdf"), plot = p_vis, width = 6.5, height = 3, dpi = 300)
```

```
# ddi lag1
```

```
emi_model_ddilag1 <- lm(emi ~ year + ddi_lag1 + country, data = df_all)
summary(emi_model_ddilag1)
```

```
##
## Call:
## lm(formula = emi ~ year + ddi_lag1 + country, data = df_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.242335 -0.050557  0.001301  0.050756  0.209920
##
```

```

## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.420083   0.065047  -6.458 4.42e-10 ***
## year1948       0.111358   0.074361   1.498 0.135339
## year1949       0.085107   0.074365   1.144 0.253372
## year1950       0.060498   0.069837   0.866 0.387047
## year1951       0.062312   0.069952   0.891 0.373785
## year1952       0.099128   0.069996   1.416 0.157784
## year1953       0.053201   0.069959   0.760 0.447593
## year1954       0.153005   0.070009   2.185 0.029647 *
## year1955       0.175182   0.070147   2.497 0.013063 *
## year1956       0.178958   0.070160   2.551 0.011261 *
## year1957       0.187335   0.070192   2.669 0.008036 **
## year1958       0.218372   0.070216   3.110 0.002056 **
## year1959       0.217400   0.070252   3.095 0.002162 **
## year1960       0.206433   0.070280   2.937 0.003575 **
## year1961       0.237508   0.070435   3.372 0.000847 ***
## year1962       0.293523   0.070505   4.163 4.13e-05 ***
## year1963       0.322364   0.070554   4.569 7.24e-06 ***
## year1964       0.280403   0.070631   3.970 9.06e-05 ***
## year1965       0.334025   0.070648   4.728 3.53e-06 ***
## year1966       0.364055   0.070669   5.152 4.76e-07 ***
## year1967       0.347576   0.070733   4.914 1.49e-06 ***
## year1968       0.277836   0.070769   3.926 0.000108 ***
## year1969       0.321939   0.070751   4.550 7.87e-06 ***
## year1970       0.271973   0.070780   3.843 0.000149 ***
## year1971       0.313138   0.071037   4.408 1.47e-05 ***
## year1972       0.295634   0.071061   4.160 4.18e-05 ***
## year1973       0.335763   0.071109   4.722 3.63e-06 ***
## year1974       0.281839   0.071163   3.960 9.41e-05 ***
## year1975       0.295248   0.071192   4.147 4.42e-05 ***
## year1976       0.292055   0.071369   4.092 5.53e-05 ***
## year1977       0.298880   0.071517   4.179 3.87e-05 ***
## year1978       0.280694   0.071752   3.912 0.000114 ***
## year1979       0.339930   0.071796   4.735 3.43e-06 ***
## year1980       0.315490   0.071801   4.394 1.56e-05 ***
## year1981       0.297777   0.072093   4.130 4.73e-05 ***
## year1982       0.243493   0.072056   3.379 0.000826 ***
## year1983       0.258018   0.068160   3.786 0.000186 ***
## year1984       0.243400   0.068649   3.546 0.000456 ***
## year1985       0.247932   0.069490   3.568 0.000420 ***
## year1986       0.223851   0.069490   3.221 0.001420 **
## year1987       0.225362   0.069549   3.240 0.001332 **
## year1988       0.237718   0.069613   3.415 0.000729 ***
## year1989       0.282985   0.069795   4.055 6.45e-05 ***
## year1990       0.273377   0.072948   3.748 0.000215 ***
## year1991       0.139698   0.068499   2.039 0.042311 *
## year1992       0.159614   0.069123   2.309 0.021635 *
## year1993       0.159532   0.069379   2.299 0.022185 *
## year1994       0.185270   0.069518   2.665 0.008126 **
## year1995       0.145274   0.069489   2.091 0.037429 *
## year1996       0.158136   0.069403   2.279 0.023418 *
## year1997       0.210404   0.069460   3.029 0.002672 **
## year1998       0.214875   0.069523   3.091 0.002190 **

```

```
## year1999      0.215669  0.069523  3.102 0.002109 **
## year2000      0.279237  0.069557  4.014 7.58e-05 ***
## year2001      0.235573  0.069548  3.387 0.000803 ***
## year2002      0.233777  0.069123  3.382 0.000818 ***
## year2003      0.264980  0.069264  3.826 0.000160 ***
## year2004      0.265985  0.069470  3.829 0.000158 ***
## year2005      0.240558  0.069582  3.457 0.000627 ***
## year2006      0.236566  0.069523  3.403 0.000760 ***
## year2007      0.250074  0.069509  3.598 0.000377 ***
## year2008      0.298450  0.069675  4.283 2.50e-05 ***
## year2009      0.256751  0.069680  3.685 0.000273 ***
## year2010      0.263432  0.069773  3.776 0.000193 ***
## year2011      0.232549  0.069650  3.339 0.000951 ***
## year2012      0.222939  0.069470  3.209 0.001480 **
## year2013      0.194028  0.069393  2.796 0.005516 **
## year2014      0.220055  0.068725  3.202 0.001516 **
## year2015      0.180795  0.068426  2.642 0.008681 **
## year2016      0.195933  0.068057  2.879 0.004285 **
## year2017      0.201069  0.067095  2.997 0.002963 **
## year2018      0.180542  0.066431  2.718 0.006965 **
## year2019      0.211447  0.066421  3.183 0.001613 **
## year2020      0.172063  0.066360  2.593 0.009998 **
## year2021      0.184270  0.066084  2.788 0.005645 **
## year2022      0.226279  0.067973  3.329 0.000983 ***
## year2023      0.146928  0.075700  1.941 0.053231 .
## year2024      0.144343  0.084627  1.706 0.089140 .
## year2025      0.004437  0.085807  0.052 0.958798
## ddi_lag1      0.265280  0.078433  3.382 0.000817 ***
## countryGermany -0.029287  0.021492 -1.363 0.174027
## countryIceland -0.011574  0.014612 -0.792 0.428932
## countryItaly   0.029933  0.014678  2.039 0.042322 *
## countryPoland  0.033273  0.021177  1.571 0.117218
## countryTurkey  0.118943  0.038732  3.071 0.002335 **
## countryWest Germany -0.037169  0.021442 -1.734 0.084061 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09089 on 292 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.4979, Adjusted R-squared:  0.3517
## F-statistic: 3.406 on 85 and 292 DF, p-value: 6.025e-15
```

```
vif(emi_model_ddilag1)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## year          4.272726 78          1.009353
## ddi_lag1      7.421423  1          2.724229
## country      10.844189  6          1.219738
```

```
adf.test(residuals(emi_model_ddilag1))
```

```
## Warning in adf.test(residuals(emi_model_ddilag1)): p-value smaller than printed
## p-value
```

```

##
## Augmented Dickey-Fuller Test
##
## data: residuals(emi_model_ddilag1)
## Dickey-Fuller = -5.6683, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary

kpss.test(residuals(emi_model_ddilag1))

## Warning in kpss.test(residuals(emi_model_ddilag1)): p-value greater than
## printed p-value

##
## KPSS Test for Level Stationarity
##
## data: residuals(emi_model_ddilag1)
## KPSS Level = 0.070173, Truncation lag parameter = 5, p-value = 0.1

jarque.bera.test(residuals(emi_model_ddilag1))

##
## Jarque Bera Test
##
## data: residuals(emi_model_ddilag1)
## X-squared = 0.53382, df = 2, p-value = 0.7657

p_vis <- visreg(emi_model_ddilag1, "ddi_lag1", gg = TRUE, xlab="DDI(previous year)", ylab="EMI", line.
  theme_house +
  scale_color_manual(values = "black") +
  scale_fill_manual(values = "grey70")

ggsave(filename = file.path(output_path, "EMI-DDIprevious-model.pdf"), plot = p_vis, width = 6.5, height = 4.5)

modelsummary(
  list(
    "EMI" = emi_model,
    "EMI" = emi_model_ddilag1
  ),
  coef_omit = "year|country",
  gof_omit="AIC|BIC|RMSE|Log.Lik.",
  output = "output/EMI-DDI-LM_summary.html",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}", conf_level=0.95,
  notes = "Values in square brackets represent 95% confidence intervals."
)

tex <- modelsummary(
  list(
    "EMI" = emi_model,
    "EMI" = emi_model_ddilag1
  ),

```

```

coef_omit = "year|country",
gof_omit="AIC|BIC|RMSE|Log.Lik.",
output = "tinytable",
fmt = NULL,
estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
statistic = "p={f_p(p.value)}", conf_level=0.95,
notes = "Values in square brackets represent 95% confidence intervals."
)|>
  theme_tt("tabular")
  save_tt(tex, "./output/EMI-DDI-LM_summary.tex", overwrite = TRUE)

ddi_model <- lm(ddi ~ year + emi + country , data = df_all)
summary(ddi_model)

```

```

##
## Call:
## lm(formula = ddi ~ year + emi + country, data = df_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.210708 -0.031154 -0.004399  0.038990  0.153859
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.522924   0.041329  12.653 < 2e-16 ***
## year1947       0.060050   0.055133   1.089 0.276955
## year1948       0.040328   0.055394   0.728 0.467173
## year1949       0.077435   0.051677   1.498 0.135079
## year1950       0.083377   0.051785   1.610 0.108440
## year1951       0.087460   0.051801   1.688 0.092385 .
## year1952       0.077104   0.051942   1.484 0.138753
## year1953       0.090387   0.051776   1.746 0.081886 .
## year1954       0.088416   0.052225   1.693 0.091504 .
## year1955       0.085510   0.052394   1.632 0.103722
## year1956       0.087853   0.052423   1.676 0.094815 .
## year1957       0.088635   0.052489   1.689 0.092335 .
## year1958       0.086820   0.052737   1.646 0.100761
## year1959       0.089338   0.052737   1.694 0.091301 .
## year1960       0.103987   0.052652   1.975 0.049192 *
## year1961       0.103961   0.052948   1.963 0.050524 .
## year1962       0.098510   0.053523   1.841 0.066684 .
## year1963       0.099484   0.053860   1.847 0.065728 .
## year1964       0.107207   0.053409   2.007 0.045623 *
## year1965       0.100070   0.054022   1.852 0.064959 .
## year1966       0.099699   0.054406   1.833 0.067873 .
## year1967       0.104646   0.054210   1.930 0.054507 .
## year1968       0.114454   0.053409   2.143 0.032926 *
## year1969       0.109447   0.053900   2.031 0.043188 *
## year1970       0.134111   0.053350   2.514 0.012471 *
## year1971       0.128310   0.053855   2.383 0.017822 *
## year1972       0.134048   0.053658   2.498 0.013022 *
## year1973       0.130747   0.054143   2.415 0.016344 *
## year1974       0.140991   0.053524   2.634 0.008875 **

```

## year1975	0.149020	0.053678	2.776	0.005848	**
## year1976	0.157346	0.053672	2.932	0.003633	**
## year1977	0.168403	0.053775	3.132	0.001911	**
## year1978	0.173033	0.053607	3.228	0.001386	**
## year1979	0.163705	0.054324	3.013	0.002804	**
## year1980	0.182107	0.054019	3.371	0.000847	***
## year1981	0.182577	0.053854	3.390	0.000792	***
## year1982	0.137664	0.051192	2.689	0.007566	**
## year1983	0.166034	0.051101	3.249	0.001290	**
## year1984	0.207428	0.051023	4.065	6.14e-05	***
## year1985	0.204995	0.051183	4.005	7.83e-05	***
## year1986	0.211450	0.050933	4.152	4.32e-05	***
## year1987	0.213898	0.050955	4.198	3.56e-05	***
## year1988	0.219601	0.051090	4.298	2.33e-05	***
## year1989	0.211223	0.051622	4.092	5.52e-05	***
## year1990	0.242974	0.049582	4.900	1.57e-06	***
## year1991	0.279196	0.048823	5.719	2.61e-08	***
## year1992	0.284031	0.049037	5.792	1.76e-08	***
## year1993	0.288495	0.049057	5.881	1.09e-08	***
## year1994	0.283170	0.049306	5.743	2.29e-08	***
## year1995	0.286615	0.048944	5.856	1.25e-08	***
## year1996	0.286683	0.049047	5.845	1.33e-08	***
## year1997	0.280398	0.049554	5.658	3.59e-08	***
## year1998	0.279590	0.049607	5.636	4.03e-08	***
## year1999	0.280629	0.049616	5.656	3.63e-08	***
## year2000	0.270070	0.050362	5.363	1.65e-07	***
## year2001	0.262241	0.049837	5.262	2.73e-07	***
## year2002	0.268159	0.049773	5.388	1.45e-07	***
## year2003	0.270118	0.050152	5.386	1.46e-07	***
## year2004	0.273486	0.050187	5.449	1.06e-07	***
## year2005	0.275394	0.049897	5.519	7.41e-08	***
## year2006	0.275618	0.049846	5.529	7.03e-08	***
## year2007	0.279143	0.050000	5.583	5.33e-08	***
## year2008	0.271325	0.050626	5.359	1.67e-07	***
## year2009	0.281160	0.050098	5.612	4.57e-08	***
## year2010	0.275789	0.050189	5.495	8.39e-08	***
## year2011	0.274743	0.049813	5.515	7.55e-08	***
## year2012	0.273877	0.049689	5.512	7.69e-08	***
## year2013	0.254618	0.049381	5.156	4.60e-07	***
## year2014	0.240138	0.049582	4.843	2.06e-06	***
## year2015	0.232404	0.049165	4.727	3.52e-06	***
## year2016	0.188930	0.049271	3.835	0.000154	***
## year2017	0.156544	0.049215	3.181	0.001623	**
## year2018	0.160745	0.048955	3.284	0.001147	**
## year2019	0.152486	0.049228	3.098	0.002137	**
## year2020	0.143099	0.048877	2.928	0.003678	**
## year2021	0.150650	0.048943	3.078	0.002277	**
## year2022	0.125696	0.050777	2.475	0.013862	*
## year2023	0.176608	0.055943	3.157	0.001758	**
## year2024	0.217123	0.062594	3.469	0.000600	***
## year2025	0.128207	0.062149	2.063	0.039989	*
## emi	0.160083	0.041590	3.849	0.000145	***
## countryGermany	0.090893	0.014983	6.066	3.96e-09	***
## countryIceland	-0.001054	0.010795	-0.098	0.922285	

```
## countryItaly      0.005999   0.010897   0.551 0.582368
## countryPoland     -0.097843   0.014353  -6.817 5.17e-11 ***
## countryTurkey     -0.435022   0.013580 -32.035 < 2e-16 ***
## countryWest Germany 0.139132   0.013625  10.212 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06752 on 298 degrees of freedom
## Multiple R-squared:  0.8695, Adjusted R-squared:  0.8318
## F-statistic: 23.08 on 86 and 298 DF,  p-value: < 2.2e-16
```

```
vif(ddi_model)
```

```
##          GVIF Df GVIF^(1/(2*Df))
## year      3.268593 79      1.007524
## emi       2.022887  1      1.422282
## country  1.700939  6      1.045259
```

```
adf.test(residuals(ddi_model))
```

```
## Warning in adf.test(residuals(ddi_model)): p-value smaller than printed p-value
```

```
##
## Augmented Dickey-Fuller Test
##
## data: residuals(ddi_model)
## Dickey-Fuller = -4.7715, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

```
kpss.test(residuals(ddi_model))
```

```
## Warning in kpss.test(residuals(ddi_model)): p-value greater than printed
## p-value
```

```
##
## KPSS Test for Level Stationarity
##
## data: residuals(ddi_model)
## KPSS Level = 0.074592, Truncation lag parameter = 5, p-value = 0.1
```

```
jarque.bera.test(residuals(ddi_model))
```

```
##
## Jarque Bera Test
##
## data: residuals(ddi_model)
## X-squared = 5.2748, df = 2, p-value = 0.07155
```

```

p_vis <- visreg(ddi_model, "emi", gg = TRUE, xlab="EMI", ylab="DDI", line.par = list(col = "black"))+
  theme_house +
  scale_color_manual(values = "black") +
  scale_fill_manual(values = "grey70")

ggsave(filename = file.path(output_path, "DDI-EMI-model.pdf"), plot = p_vis, width = 6.5, height = 3, dpi = 300)

#emi lag1
ddi_model_emilag1 <- lm(ddi ~ year + emi_lag1 + country, data = df_all)
summary(ddi_model_emilag1)

```

```

##
## Call:
## lm(formula = ddi ~ year + emi_lag1 + country, data = df_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.186539 -0.030486 -0.005388  0.035666  0.140089
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.5694187   0.0405124   14.055 < 2e-16 ***
## year1948        0.0004622   0.0539561    0.009  0.993171
## year1949       -0.0002531   0.0542259   -0.005  0.996279
## year1950        0.0338311   0.0505762    0.669  0.504080
## year1951        0.0324740   0.0506859    0.641  0.522225
## year1952        0.0275624   0.0507033    0.544  0.587130
## year1953        0.0289055   0.0508479    0.568  0.570153
## year1954        0.0485921   0.0506767    0.959  0.338420
## year1955        0.0381346   0.0511389    0.746  0.456445
## year1956        0.0381355   0.0513130    0.743  0.457962
## year1957        0.0399109   0.0513426    0.777  0.437585
## year1958        0.0421011   0.0514107    0.819  0.413502
## year1959        0.0409462   0.0516668    0.793  0.428711
## year1960        0.0539589   0.0516659    1.044  0.297173
## year1961        0.0606493   0.0515789    1.176  0.240611
## year1962        0.0604105   0.0518839    1.164  0.245237
## year1963        0.0595200   0.0524759    1.134  0.257626
## year1964        0.0573229   0.0528238    1.085  0.278741
## year1965        0.0634948   0.0523589    1.213  0.226231
## year1966        0.0617610   0.0529903    1.166  0.244762
## year1967        0.0607453   0.0533852    1.138  0.256109
## year1968        0.0612614   0.0531834    1.152  0.250309
## year1969        0.0712425   0.0523591    1.361  0.174673
## year1970        0.0829352   0.0528646    1.569  0.117772
## year1971        0.0901469   0.0522980    1.724  0.085818 .
## year1972        0.0878774   0.0528181    1.664  0.097230 .
## year1973        0.0931037   0.0526155    1.770  0.077852 .
## year1974        0.0901254   0.0531149    1.697  0.090800 .
## year1975        0.1065061   0.0524772    2.030  0.043307 *
## year1976        0.1131531   0.0526357    2.150  0.032397 *
## year1977        0.1257079   0.0526302    2.389  0.017552 *
## year1978        0.1269165   0.0527361    2.407  0.016721 *

```

## year1979	0.1288845	0.0525626	2.452	0.014791	*
## year1980	0.1364716	0.0533014	2.560	0.010959	*
## year1981	0.1375376	0.0529874	2.596	0.009918	**
## year1982	0.1381398	0.0528168	2.615	0.009374	**
## year1983	0.1177458	0.0501890	2.346	0.019642	*
## year1984	0.1589350	0.0500950	3.173	0.001671	**
## year1985	0.1597964	0.0500151	3.195	0.001552	**
## year1986	0.1606406	0.0501799	3.201	0.001519	**
## year1987	0.1662229	0.0499222	3.330	0.000981	***
## year1988	0.1737687	0.0499450	3.479	0.000580	***
## year1989	0.1714552	0.0500837	3.423	0.000707	***
## year1990	0.1950717	0.0529664	3.683	0.000275	***
## year1991	0.2172858	0.0486124	4.470	1.12e-05	***
## year1992	0.2358287	0.0478301	4.931	1.38e-06	***
## year1993	0.2376560	0.0480511	4.946	1.28e-06	***
## year1994	0.2363895	0.0480718	4.917	1.47e-06	***
## year1995	0.2302677	0.0483284	4.765	2.99e-06	***
## year1996	0.2369194	0.0479554	4.940	1.31e-06	***
## year1997	0.2376920	0.0480613	4.946	1.28e-06	***
## year1998	0.2315916	0.0485837	4.767	2.96e-06	***
## year1999	0.2321753	0.0486386	4.773	2.87e-06	***
## year2000	0.2317503	0.0486473	4.764	3.00e-06	***
## year2001	0.2095366	0.0494166	4.240	3.00e-05	***
## year2002	0.2195917	0.0488752	4.493	1.01e-05	***
## year2003	0.2274205	0.0488089	4.659	4.83e-06	***
## year2004	0.2274954	0.0491994	4.624	5.66e-06	***
## year2005	0.2251597	0.0492360	4.573	7.11e-06	***
## year2006	0.2274800	0.0489374	4.648	5.07e-06	***
## year2007	0.2336692	0.0488842	4.780	2.78e-06	***
## year2008	0.2322905	0.0490434	4.736	3.40e-06	***
## year2009	0.2296942	0.0496883	4.623	5.69e-06	***
## year2010	0.2303403	0.0491442	4.687	4.26e-06	***
## year2011	0.2233046	0.0492384	4.535	8.41e-06	***
## year2012	0.2243338	0.0488508	4.592	6.53e-06	***
## year2013	0.2016332	0.0487226	4.138	4.58e-05	***
## year2014	0.1937220	0.0484051	4.002	7.97e-05	***
## year2015	0.1769504	0.0486126	3.640	0.000322	***
## year2016	0.1401672	0.0481829	2.909	0.003904	**
## year2017	0.1055292	0.0482917	2.185	0.029665	*
## year2018	0.1057130	0.0482341	2.192	0.029193	*
## year2019	0.1057729	0.0479662	2.205	0.028223	*
## year2020	0.0863841	0.0482477	1.790	0.074420	.
## year2021	0.0998701	0.0478861	2.086	0.037885	*
## year2022	0.0771232	0.0496494	1.553	0.121421	
## year2023	0.1171351	0.0552668	2.119	0.034896	*
## year2024	0.1720063	0.0609824	2.821	0.005122	**
## year2025	0.0560570	0.0612971	0.915	0.361202	
## emi_lag1	0.1155389	0.0418400	2.761	0.006119	**
## countryGermany	0.0922344	0.0148696	6.203	1.89e-09	***
## countryIceland	-0.0027850	0.0106472	-0.262	0.793833	
## countryItaly	0.0106460	0.0107153	0.994	0.321270	
## countryPoland	-0.0936744	0.0142302	-6.583	2.14e-10	***
## countryTurkey	-0.4276066	0.0134411	-31.813	< 2e-16	***
## countryWest Germany	0.1370396	0.0134882	10.160	< 2e-16	***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06608 on 292 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.8693, Adjusted R-squared:  0.8313
## F-statistic: 22.85 on 85 and 292 DF,  p-value: < 2.2e-16
```

```
vif(ddi_model_emilag1)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## year      3.343717 78      1.007768
## emi_lag1  2.050871  1      1.432086
## country   1.714331  6      1.045943
```

```
adf.test(residuals(ddi_model_emilag1))
```

```
## Warning in adf.test(residuals(ddi_model_emilag1)): p-value smaller than printed
## p-value
```

```
##
## Augmented Dickey-Fuller Test
##
## data: residuals(ddi_model_emilag1)
## Dickey-Fuller = -5.2989, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

```
kpss.test(residuals(ddi_model_emilag1))
```

```
## Warning in kpss.test(residuals(ddi_model_emilag1)): p-value greater than
## printed p-value
```

```
##
## KPSS Test for Level Stationarity
##
## data: residuals(ddi_model_emilag1)
## KPSS Level = 0.071582, Truncation lag parameter = 5, p-value = 0.1
```

```
jarque.bera.test(residuals(ddi_model_emilag1))
```

```
##
## Jarque Bera Test
##
## data: residuals(ddi_model_emilag1)
## X-squared = 3.273, df = 2, p-value = 0.1947
```

```
p_vis <- visreg(ddi_model_emilag1, "emi_lag1", gg = TRUE, xlab="EMI(previous year)", ylab="DDI", line.
  theme_house +
  scale_color_manual(values = "black") +
  scale_fill_manual(values = "grey70")
```

```

ggsave(filename = file.path(output_path, "DDI-EMIprevious-model.pdf"), plot = p_vis, width = 6.5, height = 4.5)

modelsummary(
  list(
    "DDI" = ddi_model,
    "DDI" = ddi_model_emilag1
  ),
  coef_omit = "year|country",
  gof_omit="AIC|BIC|RMSE|Log.Lik.",
  output = "output/DDI-EMI-LM_summary.html",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}",
  conf_level=0.95,
  notes = "Values in square brackets represent 95% confidence intervals."
)

tex <- modelsummary(
  list(
    "DDI" = ddi_model,
    "DDI" = ddi_model_emilag1
  ),
  coef_omit = "year|country",
  gof_omit="AIC|BIC|RMSE|Log.Lik.",
  output = "tinytable",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}", conf_level=0.95,
  notes = "Values in square brackets represent 95% confidence intervals."
)|>
  theme_tt("tabular")
  save_tt(tex, "output/DDI-EMI-LM_summary.tex", overwrite = TRUE)

#transparent laws

# tpl baseline model
tpl_model_ddi_noemi <- lm(transparent_laws ~ year + country
                          + clientelism_rev + judiciary_index + log_gdp + ddi, data = df_all)
summary(tpl_model_ddi_noemi)

```

```

##
## Call:
## lm(formula = transparent_laws ~ year + country + clientelism_rev +
##     judiciary_index + log_gdp + ddi, data = df_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49018 -0.09033 -0.00850  0.08765  0.51968
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.701794   0.686216   5.395 1.43e-07 ***
## year1947       0.046253   0.135126   0.342  0.73237

```

## year1948	0.040592	0.135117	0.300	0.76407
## year1949	-0.023916	0.126636	-0.189	0.85034
## year1950	-0.063000	0.127506	-0.494	0.62162
## year1951	-0.008927	0.128402	-0.070	0.94462
## year1952	0.017196	0.128947	0.133	0.89401
## year1953	0.030054	0.130125	0.231	0.81751
## year1954	0.014006	0.131219	0.107	0.91507
## year1955	0.037695	0.132590	0.284	0.77638
## year1956	0.021717	0.132773	0.164	0.87018
## year1957	0.021735	0.133199	0.163	0.87049
## year1958	0.023655	0.133903	0.177	0.85990
## year1959	0.039067	0.135340	0.289	0.77305
## year1960	0.018287	0.136582	0.134	0.89358
## year1961	0.039917	0.137331	0.291	0.77152
## year1962	0.048758	0.138931	0.351	0.72588
## year1963	0.059899	0.140962	0.425	0.67120
## year1964	0.073442	0.142594	0.515	0.60691
## year1965	0.082521	0.143920	0.573	0.56683
## year1966	0.168769	0.145624	1.159	0.24743
## year1967	0.165787	0.146088	1.135	0.25738
## year1968	0.201840	0.146671	1.376	0.16983
## year1969	0.205793	0.148021	1.390	0.16550
## year1970	0.182727	0.150228	1.216	0.22484
## year1971	0.222709	0.152123	1.464	0.14427
## year1972	0.230894	0.153873	1.501	0.13456
## year1973	0.223293	0.156221	1.429	0.15398
## year1974	0.262274	0.157038	1.670	0.09597 .
## year1975	0.236935	0.157183	1.507	0.13280
## year1976	0.236388	0.159810	1.479	0.14017
## year1977	0.227311	0.162594	1.398	0.16317
## year1978	0.235462	0.164482	1.432	0.15335
## year1979	0.249400	0.166349	1.499	0.13489
## year1980	0.259644	0.168854	1.538	0.12521
## year1981	0.261751	0.168950	1.549	0.12240
## year1982	0.228368	0.163369	1.398	0.16322
## year1983	0.170354	0.162958	1.045	0.29671
## year1984	0.133460	0.165403	0.807	0.42040
## year1985	0.140707	0.166589	0.845	0.39901
## year1986	0.169491	0.168627	1.005	0.31567
## year1987	0.184615	0.171269	1.078	0.28196
## year1988	0.175906	0.172862	1.018	0.30971
## year1989	0.203425	0.175179	1.161	0.24650
## year1990	0.391157	0.171622	2.279	0.02338 *
## year1991	0.349354	0.173691	2.011	0.04521 *
## year1992	0.338031	0.175344	1.928	0.05485 .
## year1993	0.340390	0.177232	1.921	0.05576 .
## year1994	0.346499	0.177862	1.948	0.05236 .
## year1995	0.421040	0.179189	2.350	0.01946 *
## year1996	0.424895	0.181080	2.346	0.01962 *
## year1997	0.429249	0.183516	2.339	0.02001 *
## year1998	0.453191	0.186428	2.431	0.01566 *
## year1999	0.474611	0.187070	2.537	0.01170 *
## year2000	0.484413	0.189009	2.563	0.01088 *
## year2001	0.513763	0.187829	2.735	0.00662 **

```
## year2002      0.595570    0.189440    3.144  0.00184 **
## year2003      0.592499    0.191524    3.094  0.00217 **
## year2004      0.598247    0.194222    3.080  0.00227 **
## year2005      0.530840    0.196176    2.706  0.00721 **
## year2006      0.550906    0.199005    2.768  0.00600 **
## year2007      0.543397    0.201501    2.697  0.00741 **
## year2008      0.543025    0.200609    2.707  0.00719 **
## year2009      0.536058    0.198769    2.697  0.00741 **
## year2010      0.524418    0.200783    2.612  0.00947 **
## year2011      0.553650    0.202613    2.733  0.00667 **
## year2012      0.550596    0.202041    2.725  0.00682 **
## year2013      0.552504    0.200746    2.752  0.00629 **
## year2014      0.578678    0.200841    2.881  0.00426 **
## year2015      0.592266    0.201117    2.945  0.00349 **
## year2016      0.595226    0.200682    2.966  0.00327 **
## year2017      0.585119    0.202845    2.885  0.00421 **
## year2018      0.597350    0.204850    2.916  0.00382 **
## year2019      0.625106    0.205280    3.045  0.00254 **
## year2020      0.603768    0.200520    3.011  0.00283 **
## year2021      0.643353    0.203610    3.160  0.00175 **
## year2022      0.638137    0.207889    3.070  0.00235 **
## countryGermany 0.217702    0.046580    4.674  4.53e-06 ***
## countryIceland -0.094475    0.036867   -2.563  0.01089 *
## countryItaly   -0.320190    0.053601   -5.974  6.76e-09 ***
## countryPoland  -0.629382    0.096722   -6.507  3.34e-10 ***
## countryTurkey  -1.209776    0.150541   -8.036  2.34e-14 ***
## countryWest Germany 0.370309    0.056929    6.505  3.39e-10 ***
## clientelism_rev -0.907028    0.242435   -3.741  0.00022 ***
## judiciary_index 0.776752    0.162660    4.775  2.85e-06 ***
## log_gdp        -0.284038    0.071151   -3.992  8.30e-05 ***
## ddi            1.958688    0.223538    8.762  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1639 on 291 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.9459, Adjusted R-squared:  0.9299
## F-statistic: 59.15 on 86 and 291 DF, p-value: < 2.2e-16
```

```
vif(tpl_model_ddi_noemi)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## year          80.896763 76      1.029324
## country       555.467100  6      1.693252
## clientelism_rev 14.565899  1      3.816530
## judiciary_index  6.640297  1      2.576877
## log_gdp       24.250264  1      4.924456
## ddi           19.133583  1      4.374195
```

```
adf.test(residuals(tpl_model_ddi_noemi))
```

```
## Warning in adf.test(residuals(tpl_model_ddi_noemi)): p-value smaller than
## printed p-value
```

```
##
## Augmented Dickey-Fuller Test
##
## data: residuals(tpl_model_ddi_noemi)
## Dickey-Fuller = -4.4559, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary

kpss.test(residuals(tpl_model_ddi_noemi))

## Warning in kpss.test(residuals(tpl_model_ddi_noemi)): p-value greater than
## printed p-value

##
## KPSS Test for Level Stationarity
##
## data: residuals(tpl_model_ddi_noemi)
## KPSS Level = 0.13441, Truncation lag parameter = 5, p-value = 0.1

jarque.bera.test(residuals(tpl_model_ddi_noemi))

##
## Jarque Bera Test
##
## data: residuals(tpl_model_ddi_noemi)
## X-squared = 23.88, df = 2, p-value = 6.523e-06

#emi sig
tpl_model_ddi <- lm(transparent_laws ~ year + emi + country
                    + clientelism_rev + judiciary_index + log_gdp + ddi, data = df_all)
summary(tpl_model_ddi)

##
## Call:
## lm(formula = transparent_laws ~ year + emi + country + clientelism_rev +
##     judiciary_index + log_gdp + ddi, data = df_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.53432 -0.08567 -0.00250  0.08162  0.52378
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.320936   0.686271   6.296 1.13e-09 ***
## year1947        0.062202   0.131773   0.472 0.637254
## year1948        0.003135   0.132032   0.024 0.981073
## year1949       -0.022844   0.123438  -0.185 0.853309
## year1950       -0.080036   0.124358  -0.644 0.520346
## year1951       -0.022669   0.125206  -0.181 0.856450
## year1952       -0.009577   0.125866  -0.076 0.939400
## year1953        0.028605   0.126840   0.226 0.821733
## year1954       -0.028442   0.128338  -0.222 0.824767
```

## year1955	-0.013139	0.129855	-0.101	0.919477	
## year1956	-0.027649	0.129997	-0.213	0.831722	
## year1957	-0.030721	0.130485	-0.235	0.814035	
## year1958	-0.040720	0.131494	-0.310	0.757030	
## year1959	-0.021082	0.132763	-0.159	0.873942	
## year1960	-0.035702	0.133804	-0.267	0.789796	
## year1961	-0.027984	0.134917	-0.207	0.835830	
## year1962	-0.040681	0.137226	-0.296	0.767095	
## year1963	-0.037673	0.139515	-0.270	0.787332	
## year1964	-0.003603	0.140299	-0.026	0.979528	
## year1965	-0.015342	0.142368	-0.108	0.914259	
## year1966	0.060361	0.144468	0.418	0.676392	
## year1967	0.064305	0.144604	0.445	0.656869	
## year1968	0.131272	0.144033	0.911	0.362842	
## year1969	0.119759	0.145850	0.821	0.412258	
## year1970	0.121998	0.147206	0.829	0.407924	
## year1971	0.145883	0.149500	0.976	0.329973	
## year1972	0.164294	0.150893	1.089	0.277141	
## year1973	0.142874	0.153576	0.930	0.352981	
## year1974	0.205795	0.153711	1.339	0.181672	
## year1975	0.174396	0.153996	1.132	0.258373	
## year1976	0.177288	0.156462	1.133	0.258106	
## year1977	0.167610	0.159178	1.053	0.293231	
## year1978	0.184647	0.160823	1.148	0.251858	
## year1979	0.175771	0.163172	1.077	0.282279	
## year1980	0.199876	0.165255	1.209	0.227458	
## year1981	0.207506	0.165232	1.256	0.210182	
## year1982	0.193148	0.159483	1.211	0.226847	
## year1983	0.136285	0.159068	0.857	0.392277	
## year1984	0.111974	0.161314	0.694	0.488155	
## year1985	0.114070	0.162516	0.702	0.483303	
## year1986	0.154701	0.164409	0.941	0.347514	
## year1987	0.171707	0.166975	1.028	0.304647	
## year1988	0.158781	0.168550	0.942	0.346956	
## year1989	0.171645	0.170937	1.004	0.316147	
## year1990	0.382730	0.167301	2.288	0.022877	*
## year1991	0.377518	0.169449	2.228	0.026653	*
## year1992	0.356423	0.170977	2.085	0.037978	*
## year1993	0.360327	0.172827	2.085	0.037953	*
## year1994	0.354992	0.173383	2.047	0.041516	*
## year1995	0.448931	0.174801	2.568	0.010722	*
## year1996	0.449963	0.176617	2.548	0.011360	*
## year1997	0.435817	0.178890	2.436	0.015443	*
## year1998	0.462124	0.181734	2.543	0.011515	*
## year1999	0.483251	0.182358	2.650	0.008491	**
## year2000	0.467656	0.184283	2.538	0.011682	*
## year2001	0.515363	0.183086	2.815	0.005214	**
## year2002	0.602593	0.184664	3.263	0.001233	**
## year2003	0.587515	0.186692	3.147	0.001821	**
## year2004	0.594637	0.189320	3.141	0.001858	**
## year2005	0.540203	0.191236	2.825	0.005059	**
## year2006	0.566393	0.194018	2.919	0.003783	**
## year2007	0.554753	0.196433	2.824	0.005070	**
## year2008	0.529553	0.195572	2.708	0.007177	**

```
## year2009      0.538842  0.193751  2.781 0.005772 **
## year2010      0.525251  0.195713  2.684 0.007698 **
## year2011      0.570171  0.197539  2.886 0.004190 **
## year2012      0.570525  0.197001  2.896 0.004066 **
## year2013      0.583835  0.195831  2.981 0.003114 **
## year2014      0.600294  0.195843  3.065 0.002380 **
## year2015      0.631161  0.196275  3.216 0.001448 **
## year2016      0.624992  0.195754  3.193 0.001564 **
## year2017      0.617135  0.197882  3.119 0.001999 **
## year2018      0.643048  0.199998  3.215 0.001450 **
## year2019      0.657946  0.200261  3.285 0.001143 **
## year2020      0.648066  0.195764  3.310 0.001049 **
## year2021      0.685411  0.198742  3.449 0.000647 ***
## year2022      0.663815  0.202739  3.274 0.001188 **
## emi           0.429718  0.106525  4.034 7.02e-05 ***
## countryGermany 0.223568  0.045427  4.922 1.44e-06 ***
## countryIceland -0.095834 0.035938 -2.667 0.008091 **
## countryItaly   -0.382284 0.054467 -7.019 1.59e-11 ***
## countryPoland  -0.702818 0.096021 -7.319 2.46e-12 ***
## countryTurkey  -1.364896 0.151694 -8.998 < 2e-16 ***
## countryWest Germany 0.399283 0.055955 7.136 7.73e-12 ***
## clientelism_rev -1.110133 0.241617 -4.595 6.47e-06 ***
## judiciary_index 0.716895 0.159245 4.502 9.77e-06 ***
## log_gdp        -0.338790 0.070670 -4.794 2.62e-06 ***
## ddi            1.943623 0.217925 8.919 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1598 on 290 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.9488, Adjusted R-squared:  0.9334
## F-statistic: 61.73 on 87 and 290 DF, p-value: < 2.2e-16
```

```
vif(tpl_model_ddi)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## year          142.103609 76      1.033146
## emi            2.263986  1      1.504655
## country       635.881301  6      1.712437
## clientelism_rev 15.227108  1      3.902193
## judiciary_index  6.698454  1      2.588137
## log_gdp        25.179065  1      5.017875
## ddi           19.139203  1      4.374837
```

```
adf.test(residuals(tpl_model_ddi))
```

```
## Warning in adf.test(residuals(tpl_model_ddi)): p-value smaller than printed
## p-value
```

```
##
## Augmented Dickey-Fuller Test
##
```

```
## data: residuals(tpl_model_ddi)
## Dickey-Fuller = -4.6267, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

```
kpss.test(residuals(tpl_model_ddi))
```

```
## Warning in kpss.test(residuals(tpl_model_ddi)): p-value greater than printed
## p-value
```

```
##
## KPSS Test for Level Stationarity
##
## data: residuals(tpl_model_ddi)
## KPSS Level = 0.12349, Truncation lag parameter = 5, p-value = 0.1
```

```
jarque.bera.test(residuals(tpl_model_ddi))
```

```
##
## Jarque Bera Test
##
## data: residuals(tpl_model_ddi)
## X-squared = 35.572, df = 2, p-value = 1.886e-08
```

```
#JB fails
```

```
bootf <- function(df, indices)
{
  tpl_model_ddi <- lm(transparent_laws ~ year + emi + country
    + clientelism_rev + judiciary_index + log_gdp + ddi, data = df,
    subset=indices
  )
  return(tpl_model_ddi$coefficients['emi'])
}
boots <- boot(df_all, bootf, R=10000)
quantile(boots$t, p=c(0.025, 0.5, 0.975)) # to verify, we perform a bootstrapping test, lower part of
```

```
##      2.5%      50%      97.5%
## 0.2072713 0.4276007 0.6492963
```

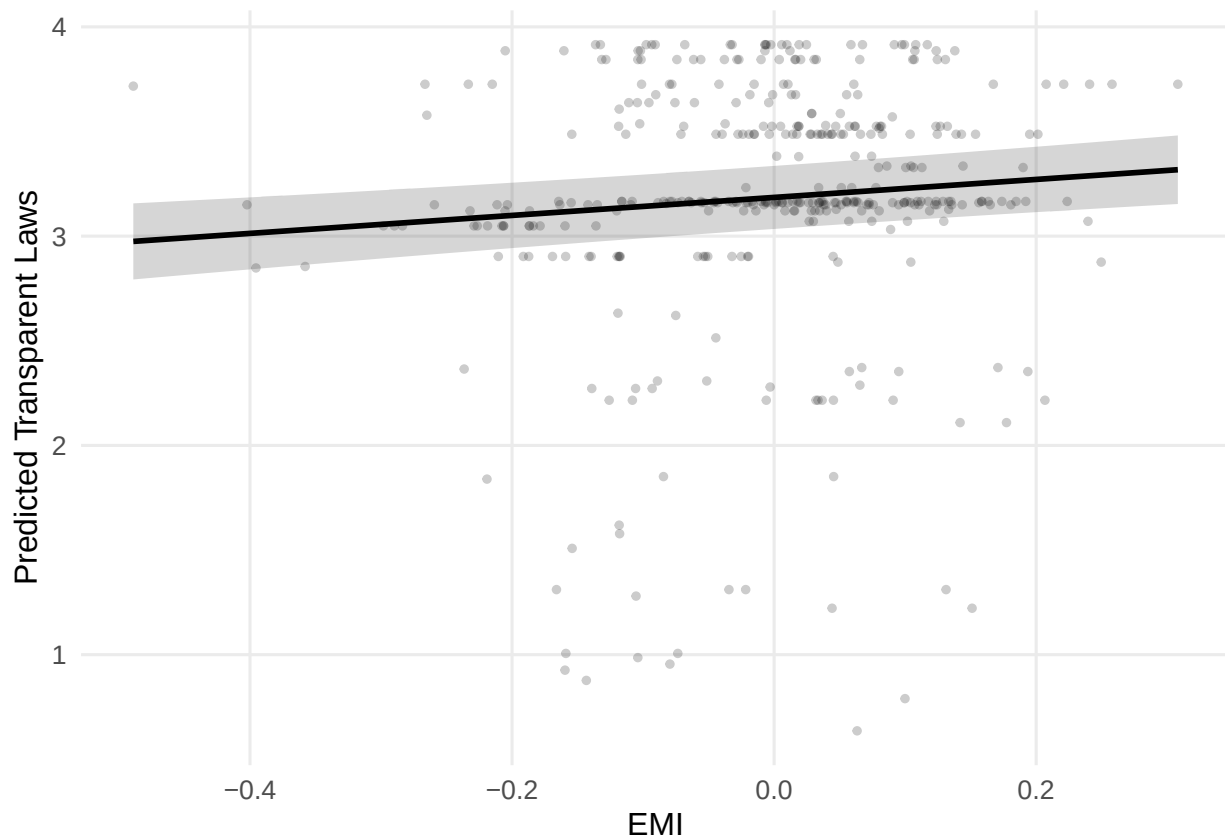
```
lrtest(tpl_model_ddi_noemi, tpl_model_ddi)
```

```
## Likelihood ratio test
##
## Model 1: transparent_laws ~ year + country + clientelism_rev + judiciary_index +
##      log_gdp + ddi
## Model 2: transparent_laws ~ year + emi + country + clientelism_rev + judiciary_index +
##      log_gdp + ddi
##      #Df LogLik Df  Chisq Pr(>Chisq)
## 1   88 196.62
## 2   89 206.94  1 20.637  5.551e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

#plot TPL model
emi_seq <- seq(min(df_all$emi, na.rm=TRUE), max(df_all$emi, na.rm=TRUE), length.out=50)
mf <- model.frame(tpl_model_ddi)
nd <- mf[rep(seq_len(nrow(mf)), each = length(emi_seq)), , drop = FALSE]
nd$emi <- rep(emi_seq, times = nrow(mf))
pred_all <- predictions(
  tpl_model_ddi,
  newdata = nd
)
pred <- aggregate(cbind(estimate, conf.low, conf.high) ~ emi, data = pred_all, mean)
ggplot() +
  geom_point(data = df_all, aes(y=transparent_laws, x=emi), alpha = 0.2, size = 1) +
  geom_ribbon(data = pred, aes(x=emi, ymin = conf.low, ymax = conf.high), alpha = 0.2) +
  geom_line(data = pred, aes(x=emi, y = estimate), linewidth = 1) +
  # theme_minimal() +
  labs(x = "EMI", y = "Predicted Transparent Laws")

```



```

ggsave(filename = file.path(output_path, "TPL-EMI-model.pdf"), width = 6.5, height = 3, dpi = 300, device = "pdf")

df_common <- df_all[
  complete.cases(df_all[, c(
    "transparent_laws", "year", "emi_lag1", "country",
    "clientelism_rev", "judiciary_index", "log_gdp", "ddi"
  ))],
]

```

```

tpl_model_ddi_emilag1_base <- lm(transparent_laws ~ year + country
                                + clientelism_rev + judiciary_index + log_gdp + ddi, data = df_common)

tpl_model_ddi_emilag1 <- lm(transparent_laws ~ year + emi_lag1 + country
                            + clientelism_rev + judiciary_index + log_gdp + ddi, data = df_common)

lrtest(tpl_model_ddi_emilag1_base, tpl_model_ddi_emilag1)

```

```

## Likelihood ratio test
##
## Model 1: transparent_laws ~ year + country + clientelism_rev + judiciary_index +
##   log_gdp + ddi
## Model 2: transparent_laws ~ year + emi_lag1 + country + clientelism_rev +
##   judiciary_index + log_gdp + ddi
##   #Df LogLik Df  Chisq Pr(>Chisq)
## 1   87 193.37
## 2   88 204.80  1 22.858  1.744e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
summary(tpl_model_ddi_emilag1)
```

```

##
## Call:
## lm(formula = transparent_laws ~ year + emi_lag1 + country + clientelism_rev +
##   judiciary_index + log_gdp + ddi, data = df_common)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.54240 -0.08124 -0.00137  0.07753  0.52048
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.179587   0.806190   5.184 4.12e-07 ***
## year1948      -0.003725   0.130062  -0.029 0.977170
## year1949      -0.100558   0.130922  -0.768 0.443082
## year1950      -0.122783   0.122047  -1.006 0.315256
## year1951      -0.094197   0.122464  -0.769 0.442428
## year1952      -0.069260   0.122759  -0.564 0.573065
## year1953      -0.073965   0.123628  -0.598 0.550125
## year1954      -0.068989   0.124100  -0.556 0.578708
## year1955      -0.092449   0.125925  -0.734 0.463458
## year1956      -0.115402   0.126476  -0.912 0.362313
## year1957      -0.117038   0.126898  -0.922 0.357154
## year1958      -0.119130   0.127592  -0.934 0.351263
## year1959      -0.117841   0.129148  -0.912 0.362306
## year1960      -0.138101   0.130336  -1.060 0.290237
## year1961      -0.111701   0.130924  -0.853 0.394283
## year1962      -0.118081   0.132913  -0.888 0.375071
## year1963      -0.133157   0.135636  -0.982 0.327073
## year1964      -0.132511   0.137750  -0.962 0.336886
## year1965      -0.104435   0.138262  -0.755 0.450674

```

## year1966	-0.042344	0.140967	-0.300	0.764104
## year1967	-0.059124	0.142124	-0.416	0.677719
## year1968	-0.015177	0.142376	-0.107	0.915185
## year1969	0.022126	0.142455	0.155	0.876680
## year1970	-0.020528	0.145307	-0.141	0.887753
## year1971	0.042414	0.146433	0.290	0.772299
## year1972	0.030022	0.148829	0.202	0.840279
## year1973	0.030642	0.150982	0.203	0.839317
## year1974	0.051370	0.152453	0.337	0.736399
## year1975	0.049906	0.151910	0.329	0.742759
## year1976	0.043270	0.154919	0.279	0.780211
## year1977	0.034064	0.157911	0.216	0.829363
## year1978	0.038565	0.160065	0.241	0.809781
## year1979	0.059788	0.161892	0.369	0.712170
## year1980	0.041587	0.165140	0.252	0.801355
## year1981	0.055381	0.165075	0.335	0.737505
## year1982	0.062314	0.164563	0.379	0.705222
## year1983	-0.018374	0.157522	-0.117	0.907223
## year1984	-0.038861	0.160704	-0.242	0.809097
## year1985	-0.027709	0.162026	-0.171	0.864331
## year1986	-0.006871	0.164370	-0.042	0.966687
## year1987	0.018489	0.167146	0.111	0.912001
## year1988	0.008511	0.168974	0.050	0.959866
## year1989	0.029424	0.171147	0.172	0.863622
## year1990	0.174879	0.180026	0.971	0.332170
## year1991	0.178530	0.169690	1.052	0.293649
## year1992	0.204459	0.171578	1.192	0.234397
## year1993	0.194706	0.173594	1.122	0.262973
## year1994	0.199815	0.174375	1.146	0.252802
## year1995	0.263405	0.175936	1.497	0.135462
## year1996	0.286922	0.178265	1.610	0.108613
## year1997	0.288379	0.181125	1.592	0.112461
## year1998	0.287957	0.184137	1.564	0.118974
## year1999	0.307487	0.184990	1.662	0.097579 .
## year2000	0.318087	0.187308	1.698	0.090564 .
## year2001	0.319217	0.186014	1.716	0.087236 .
## year2002	0.422381	0.187739	2.250	0.025226 *
## year2003	0.421358	0.190092	2.217	0.027443 *
## year2004	0.412753	0.193246	2.136	0.033543 *
## year2005	0.344712	0.195450	1.764	0.078860 .
## year2006	0.376541	0.198538	1.897	0.058900 .
## year2007	0.371082	0.201512	1.841	0.066593 .
## year2008	0.364882	0.200826	1.817	0.070286 .
## year2009	0.334671	0.198760	1.684	0.093319 .
## year2010	0.339925	0.200814	1.693	0.091601 .
## year2011	0.365566	0.202815	1.802	0.072534 .
## year2012	0.377809	0.202340	1.867	0.062904 .
## year2013	0.385735	0.200833	1.921	0.055774 .
## year2014	0.424156	0.200967	2.111	0.035683 *
## year2015	0.427887	0.201164	2.127	0.034277 *
## year2016	0.442924	0.200189	2.213	0.027725 *
## year2017	0.419973	0.201845	2.081	0.038360 *
## year2018	0.432784	0.204012	2.121	0.034757 *
## year2019	0.474001	0.204600	2.317	0.021231 *

```
## year2020      0.439055    0.199143    2.205 0.028275 *
## year2021      0.497125    0.203037    2.448 0.014952 *
## year2022      0.474279    0.207184    2.289 0.022804 *
## emi_lag1      0.456604    0.107478    4.248 2.92e-05 ***
## countryGermany 0.201728    0.046320    4.355 1.86e-05 ***
## countryIceland -0.091032    0.038814   -2.345 0.019699 *
## countryItaly   -0.364961    0.056912   -6.413 5.95e-10 ***
## countryPoland  -0.654411    0.107419   -6.092 3.61e-09 ***
## countryTurkey  -1.241211    0.172644   -7.189 5.78e-12 ***
## countryWest Germany 0.355561    0.058856    6.041 4.79e-09 ***
## clientelism_rev -0.870137    0.267059   -3.258 0.001257 **
## judiciary_index 0.591007    0.166690    3.546 0.000458 ***
## log_gdp        -0.301460    0.080345   -3.752 0.000213 ***
## ddi            2.032352    0.235596    8.626 4.56e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1592 on 284 degrees of freedom
## Multiple R-squared:  0.9474, Adjusted R-squared:  0.9315
## F-statistic: 59.52 on 86 and 284 DF,  p-value: < 2.2e-16
```

```
vif(tpl_model_ddi_emilag1)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## year          180.274437 75      1.035236
## emi_lag1       2.286639  1      1.512164
## country       917.831020  6      1.765619
## clientelism_rev 18.141714  1      4.259309
## judiciary_index  6.918991  1      2.630397
## log_gdp        30.686120  1      5.539505
## ddi           21.096369  1      4.593078
```

```
adf.test(residuals(tpl_model_ddi_emilag1))
```

```
## Warning in adf.test(residuals(tpl_model_ddi_emilag1)): p-value smaller than
## printed p-value
```

```
##
## Augmented Dickey-Fuller Test
##
## data: residuals(tpl_model_ddi_emilag1)
## Dickey-Fuller = -4.2503, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

```
kpss.test(residuals(tpl_model_ddi_emilag1))
```

```
## Warning in kpss.test(residuals(tpl_model_ddi_emilag1)): p-value greater than
## printed p-value
```

```
##
## KPSS Test for Level Stationarity
```

```
##
## data: residuals(tpl_model_ddi_emilag1)
## KPSS Level = 0.12451, Truncation lag parameter = 5, p-value = 0.1
```

```
jarque.bera.test(residuals(tpl_model_ddi_emilag1))
```

```
##
## Jarque Bera Test
##
## data: residuals(tpl_model_ddi_emilag1)
## X-squared = 50.329, df = 2, p-value = 1.178e-11
```

```
#JB fails
```

```
bootf <- function(df, indices)
```

```
{
  tpl_model_ddi_emilag1 <- lm(transparent_laws ~ year + emi_lag1 + country
                             + clientelism_rev + judiciary_index + log_gdp + ddi, data = df,
                             subset=indices
                             )
  return(tpl_model_ddi_emilag1$coefficients['emi_lag1'])
}
```

```
boots <- boot(df_common, bootf, R=10000)
```

```
quantile(boots$t, p=c(0.025, 0.5, 0.975)) # to verify, we perform a bootstrapping test, lower part of
```

```
##      2.5%      50%      97.5%
## 0.2382274 0.4601581 0.6820767
```

```
#plot emi_lag1
```

```
emi_seq <- seq(min(df_all$emi_lag1, na.rm=TRUE), max(df_all$emi_lag1, na.rm=TRUE), length.out=50)
```

```
mf <- model.frame(tpl_model_ddi_emilag1)
```

```
nd <- mf[rep(seq_len(nrow(mf)), each = length(emi_seq)), , drop = FALSE]
```

```
nd$emi_lag1 <- rep(emi_seq, times = nrow(mf))
```

```
pred_all <- predictions(
  tpl_model_ddi_emilag1,
  newdata = nd
)
```

```
pred <- aggregate(cbind(estimate, conf.low, conf.high) ~ emi_lag1, data = pred_all, mean)
```

```
ggplot() +
```

```
  geom_point(data = df_all, aes(y=transparent_laws, x=emi_lag1), alpha = 0.2, size = 1) +
```

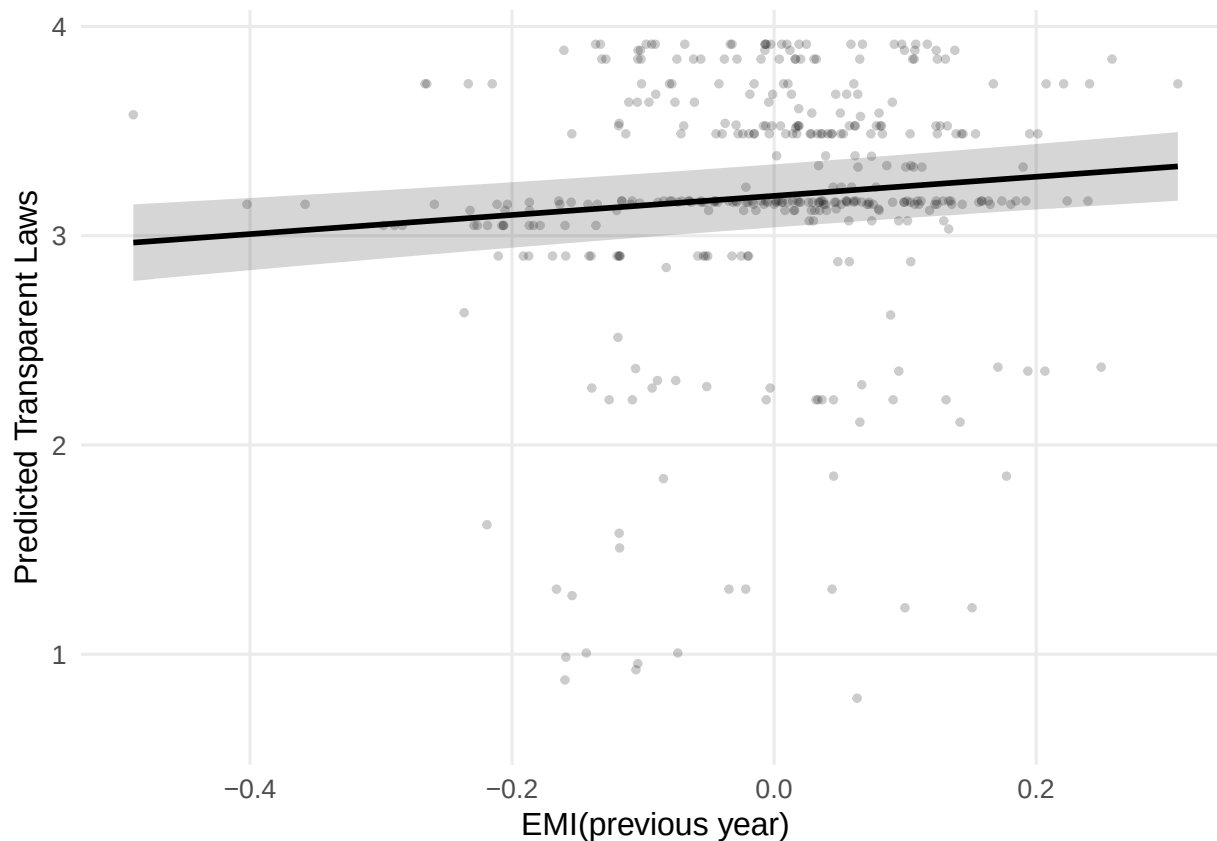
```
  geom_ribbon(data = pred, aes(x=emi_lag1, ymin = conf.low, ymax = conf.high), alpha = 0.2) +
```

```
  geom_line(data = pred, aes(x=emi_lag1, y = estimate), linewidth = 1) +
```

```
  # theme_minimal() +
```

```
  labs(x = "EMI(previous year)", y = "Predicted Transparent Laws")
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
ggsave(filename = file.path(output_path, "TPL-EMIprevious-model.pdf"), width = 6.5, height = 3, dpi = 300)
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
modelsummary(
  list(
    "TPL index" = tpl_model_ddi_noemi,
    "TPL index" = tpl_model_ddi,
    "TPL index" = tpl_model_ddi_emilag1_base,
    "TPL index" = tpl_model_ddi_emilag1
  ),
  coef_omit = "year|country",
  gof_omit="AIC|BIC|RMSE|Log.Lik.",
  output = "output/TPL-EMI-LM_summary.html",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}", conf_level=0.95,
  notes = "Values in square brackets represent 95% confidence intervals."
)

tex <- modelsummary(
  list(
    "TPL index" = tpl_model_ddi_noemi,
    "TPL index" = tpl_model_ddi,
```

```

    "TPL index" = tpl_model_ddi_emilag1_base,
    "TPL index" = tpl_model_ddi_emilag1
  ),
  coef_omit = "year|country",
  gof_omit="AIC|BIC|RMSE|Log.Lik.",
  output = "tinytable",
  fmt = NULL,
  estimate = "{f_est(estimate)} [{f_est(conf.low)}, {f_est(conf.high)}]",
  statistic = "p={f_p(p.value)}", conf_level=0.95,
  notes = "Values in square brackets represent 95% confidence intervals."
)|>
  theme_tt("tabular")
  save_tt(tex, "output/TPL-EMI-LM_summary.tex", overwrite = TRUE)

```

proc

```
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
```

```
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      cov, smooth, var
```

```
df_prolific <- read.csv("../aggregate_data/prolific_ratings_emiw2v_llms_sberr.csv")
```

```
roc_sberrllms <- roc(df_prolific$label, df_prolific$rating_emb_pool)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
roc_w2v <- roc(df_prolific$label, df_prolific$emi_w2v)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
roc.test(roc_sberrllms, roc_w2v, method = "delong")
```

```
##
```

```
## DeLong's test for two correlated ROC curves
```

```
##
```

```
## data: roc_sberrllms and roc_w2v
```

```
## Z = 2.5909, p-value = 0.009573
```

```
## alternative hypothesis: true difference in AUC is not equal to 0
## 95 percent confidence interval:
##  0.008334634 0.060116653
## sample estimates:
## AUC of roc1 AUC of roc2
##  0.8250345  0.7908088
```